

2023-Ph-H1-Q6

Section: Particles and Waves

Topic: Spectra

What is the origin of the line spectrum of hydrogen?

Answer: A

(Electrons falling from higher to lower energy levels)

Guidance for Students:

When an electron in hydrogen drops to a lower energy level, it emits a photon. The photon's energy corresponds to a specific wavelength — creating spectral lines.

Revision Tips:

- Spectral lines = transitions between energy levels.
- Energy of photon: $E = hf$
- Emission = falling down → light emitted.
- Absorption = jumping up → light absorbed.